

Exaptation as source of creativity, innovation, and diversity: introduction to the Special Section

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Abstract

Exaptation is one of the most important and, yet, little studied evolutionary mechanisms in the history of species, ecosystems, and artifacts (e.g., technologies). Many, if not most, of all biological traits and human artifacts that were developed for particular functions started as something different: feathers were most likely selected for thermal insulation, bones originated as excess calcium repositories, microwave ovens started life as radar magnetrons. Exaptation thus describes a discontinuous evolutionary process resulting from a functional shift of an existing trait or artifact. Despite the importance of the concept, even the term exaptation still remains largely unknown outside the field of evolutionary biology. The main purpose of this special section is to introduce the concept of exaptation to a broader audience, discuss its significance, and expose its contribution to the (possible) solution of long-standing, yet unresolved, questions about the emergence of novelty, particularly radical innovation.

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“Such explanations, and all other such that men give, put effect for cause and are based on perverted reasoning; since nothing is born in us simply in order that we may use it, but that which is born creates the use. There was no sight before the eyes with their light were born, no speaking of words before the tongue was made; but rather the origin of the tongue came long before speech, and the ear was made long before sound was heard, in a word all the members, as I think, existed before their use; they could not then have grown up for the sake of use” (Lucretius, *De Rerum Natura*, IV-832–842).

1. Introduction: why do we need a theory on exaptation

In March 1865, Joseph Lister performed the first of a series of surgeries that would revolutionize surgery and medicine, and for which he would gain the name of “father of modern surgery.” Lister applied an antiseptic, carbolic acid, to dress wounds and so prevent infection. By using this method, Lister reduced post-surgery mortality from about 40% down to 15% (Wootton, 2006). The antiseptic innovation is usually interpreted as an example of practice that follows theory. By reading Pasteur’s papers, Lister learned the modern theory of germs and conceived a way to block bacterial infection: “Believing Pasteur was correct in his assumptions, Lister began the search for a substance that would kill bacteria” (Gabriel, 2013: 136).

Extant explanations of the appearance of novelty, including radical innovation, assume that first new functions emerge and then new artifacts are designed to perform those functions. Arthur (2009), for instance, argues that new functions and corresponding artifacts typically originate from one of the following two channels: the discovery of a new phenomenon in science and technology, which activates the exploration of the possibilities inherent in the new phenomenon and eventually results in the design and development of new artifacts and processes; the expression of a need that drives the formulation of functions that subsequently are translated into new artifacts. The traditional interpretation of Lister's innovation fits the former pattern: the discovery of a new phenomenon—the causal role of germs in infections—triggered an exploration trajectory that led to the practice of antiseptics.

A more careful historical analysis, however, would reveal that another pivotal event had preceded Lister's discovery of Pasteur's paper at the beginning of 1865. In Lister's words:

In the course of the 1864 I was much struck with an account of the remarkable effects produced by carbolic acid upon the sewage of the town of Carlisle [North-West England] . . . Not only preventing all odors from the lands irrigated with refuse material, but as is stated destroying the entozoa [worms and similar parasites] which usually infest cattle fed upon such pastures . . . I saw that such a powerful antiseptic was peculiarly adapted for experiments . . . The applicability of carbolic acid for the treatment of compound fracture naturally occurred to me (Parker, 2013: 208).¹

Here the causality is inverted: antiseptics is not a direct consequence of a theoretical advancement but, based on Lister's chronology of events, the idea and the first experiments of carbolic acid on animals preceded his exposure to Pasteur's modern germ theory. Lister, of course, knew about germs: in fact, he was a keen microscopist and the son of an amateur scientist, Joseph Jackson Lister, who first designed microscopes free of chromatic aberration. His antiseptics method, however, originated from his intuition about the similarity between putrefaction in sewers and in open wounds—which convinced him that putrefaction and suppuration may be the same process and, therefore, share the same cause (germs). Although others had reached similar conclusions, Lister was the first to develop a new (surgical) function from this intuition. He transformed the practice of antiseptics, which before him was used to control and diminish suppuration after it had occurred (Fisher, 1977: 130) into prevention of infection. The introduction of antiseptics marked a revolutionary advancement in medicine. In a documentary on antiseptics (sponsored by the British Medical Journal), it is stated: "He [Lister] saved more lives by the introduction of this system than all the wars of the 19th century together had sacrificed" (Freeth, 2009).

In the language of this Special Section, Lister exapted an existing artifact (carbolic acid) for a function for which it had not been originally designed. The case of Lister's innovation is not simply a statement about the importance of exaptation, it also reveals that the duality "need-phenomenon" is incomplete. Exaptation, in fact, identifies a third channel driving the emergence of novelty: a functional shift of an existing artifact that is not traceable to the discovery of new phenomena or the pull from clearly defined needs. As the papers in this Special Section (in addition to a recent and expanding literature on the subject) show, the function-driven channel represents a key source of creativity, innovation, and economic well-being of a social system. Surprisingly, though, this channel is rarely mentioned explicitly and/or analyzed in traditional accounts of the origins of novelty, particularly radical innovation.

But what is exaptation? Sometimes referred to as *pre-adaptation* (Bock, 1959; Ridley, 1999), exaptation is one of the most important and, yet, little studied evolutionary mechanisms in the history of species, ecosystems, and artifacts (e.g., technologies). Gould and Vrba (1982: 6) refer to the process whereby "characters, evolved for other usages (or for no function at all), and later 'coopted' for their current role." They make a distinction between *adaptation*, i.e., the gradual modification of a trait driven by natural selection and one which assumes continuity in the selected function, and *exaptation*, i.e., a discontinuous evolutionary process expressed by functional change of a biological trait which may open a new evolutionary trajectory.

Adaptive innovation is an activity aimed to respond to a *direct* type of question: given a problem, what is the solution? In Alexander's words, the process of inventing physical things "displays new physical order, *form*, in response to *function*" (Alexander, 1964: 1, emphasis added). In processes driven by direct questions, therefore, form follows function. The form is usually a working architecture of parts and modules that performs a specific function that permits the satisfaction of a predefined need. In this special issue, we emphasize how exaptations often respond to a

1 In compound fractures bones are exposed directly to the air, generating deadly infections, for which the only remedy was amputation.

reverse type of question: “Now that I have come to a result, what problem have I solved?” (Wiener, 1993: 22). In a reverse question, the end state is largely unknown. The process starts from the assumption that forms—be they technologies, capabilities, or else—have an inherent (and unpredictable) creative potential. This creative potential consists in the appearance of novel functions for which the forms were not originally designed or selected for. In the history of technology, this potential is revealed in numerous accounts of serendipitous discoveries and technologies suddenly bifurcating into new trajectories responsible for the emergence of entirely new markets or industries (e.g., Levinthal, 1998; Meyers, 2007). In a reverse question, therefore, function follows form.

Exaptation appears to be ubiquitous. Nearly all biological traits and many commercial products developed for particular markets and functions, began their life as something different (Kauffman, 2000; Dew *et al.*, 2004; Cattani, 2005, 2006, 2008; Andriani and Carignani, 2014): feathers were most likely selected for thermal insulation, bones originated as excess calcium repositories, microwave ovens started life as radar magnetrons. The term has spread across several social science disciplines from history of technology, to technology and innovation studies (Dew *et al.*, 2004; Cattani, 2005, 2006; Bonifati, 2010; Lane, 2011; Andriani and Carignani, 2014), anthropology (Brown and Feldman, 2009), psychology (Buss *et al.*, 1998), linguistics and origin of languages (Closs Traugott, 2004), and architecture and urban studies (Keeffe *et al.*, 2012). Despite its ubiquity, exaptation has failed to become an accepted concept and analytical tool in the social sciences. But exaptation can enhance significantly our understanding of the emergence of novelty.

We are not the first to study exaptation and the problem of emergent functions. Yet, to the best of our best knowledge, this Special Section is the first ever devoted to this subject. We differ from previous work such as Padgett and Powell’s (2012) study on the emergence of new organizational forms and markets, which takes an explicit social network-centered approach. We cover some of the same ground but we put the “artifact” at the center of our analysis. Responding to Basalla’s (1988) call for the centrality of the artifact in evolutionary theories of technology and technological evolution, our artifact-centered approach provides valuable insights into the dynamics of innovation. In particular, it sheds light on the conundrum of radical innovation in which attempts to explain the simultaneous emergence of new markets and technologies must tackle complex co-evolutionary issues. By assuming that neither intentionality nor social networks are necessary to elucidate the origins of new functions, an artifact-centered approach to exaptation eschews some of those issues and provides a simpler—though by no means less useful—conceptual platform for studying the emergence of new markets and industries.

2. What do we know so far

Over the past few years, the notion of exaptation has gained traction among social scientists, especially those who are interested in technological innovation and evolution. This section briefly reviews a few select research areas in which exaptation has been studied and its importance empirically supported.

2.1 Theory of speciation

Where do new technologies come from? Several authors (Levinthal, 1998; Adner, 2002; Adner and Levinthal, 2002; Cattani, 2006) argue that the process of technological “speciation” share fundamental traits with the corresponding process of biological speciation. After reviewing the meaning of speciation in evolutionary biology, we expose the linkage between speciation and exaptation. Broadly speaking, three different processes might be at work. At one extreme, an existing species A evolves into a new species B ($A \rightarrow B$) through the gradual but steady accumulation of evolutionary novelties within a single lineage. This case is rather well understood and, in a sense, represents the baseline model of Darwinian phyletic gradualism. At the opposite extreme, a new species D comes into existence as a result of a macro-mutation. This extreme case of punctuation is consistent with Goldschmidt’s (1940) idea of *hopeful monsters*.² According to Goldschmidt, while microevolution can occur by natural selection of small genetic variants, higher groups evolve by a different process (i.e., via *hopeful monsters*). An important implication of this argument for our purposes is the uncoupling of micro- and macroevolution. In his analysis of technological change, Mokyr (1991) makes a similar sort of argument and distinguishes between macro-inventions (the analogue of macro-mutations) that create a new

2 “[...] radically new phenotypes are produced by mutations in developmental genes, and these mutations are the mechanism by which new major groups evolve” (Ridley, 1999: 600).

technique (the analogue of a new species) and micro-inventions (i.e., gradual accumulation of novelties) that refine and improve existing techniques. Without discarding the possibility that a new technique can also emerge by the accumulation of small inventions, he maintains that the term “hopeful monster” seems an apt description of such revolutionary inventions as “Gutenberg’s first printing press or Newcomen’s 1712 Dudley Castle machine” (Mokyr, 1991: 142). If macro-mutations are not the primary cause of speciation, or they are only under extraordinary circumstances, other mechanisms or evolutionary forces must be able to explain the dynamics of the process. A more frequently observed and plausible case coincides with speciation by divergent selection whereby an existing lineage splits into a new one (C), a process termed *cladogenesis* (Sober, 2000: 12–13).

While the outcome (emergence of a new species) is common to the three scenarios, the path—i.e., the particular process or mechanism prompting a speciation event—is quite different. A careful examination of the underlying micro-processes and forces is critical for understanding the genesis of novelty (e.g., a new technology). Focusing on these processes and forces would reveal how the belief that certain innovations stem from revolutionary upheavals in technology often can be attributed to a loss or concealment of crucial antecedents. As Basalla (1988: 30) puts it, “key artifacts—such as the steam engine, the cotton gin, or the transistor—emerged in an evolutionary fashion. [They] illustrate the evolutionary hypothesis despite the fact that initially they appear to be excellent candidates for use in supporting the contrary discontinuous explanation.” Extant studies on technological speciation (e.g., Levinthal, 1998; Adner, 2002; Adner and Levinthal, 2002; Cattani, 2006) emphasize how a new technology often emerges from deploying a firm’s existing technological knowledge base into a new selection environment. The selection forces (user needs and performance requirements) to which firms subject themselves often steer their R&D into novel, unanticipated directions. Focusing on the interaction with the environment sheds light on how new technologies develop and come to commercial fruition after they are invented. Extant research on technological speciation is primarily concerned with the morphological changes that occur as a result of adapting to the selection criteria of a new application domain. By describing a process that is inherently adaptive, this research tends to overlook the functional shift that this new application may entail. The same artifact can indeed be co-opted into a completely different function than the one for which it was originally selected. Even though key artifacts may emerge in a Basallian evolutionary fashion (i.e., relative morphological continuity with their antecedents), an evolutionary theory on the emergence of novelty must explicitly account for what causes the “functional shift” of an existing artifact, i.e., the functional discontinuity that is central to the notion of exaptation (Gould and Vrba, 1982).

Within the context of technology, for example, exaptation suggests that firms may accumulate technological knowledge without anticipation of its subsequent uses. Part of this technological knowledge might be functionally “pre-adapted” (i.e., valuable) for alternative, as yet unknown, applications. This is consistent with the case in which a feature or character, previously shaped by natural selection for a particular function (an adaptation), is later on co-opted for a new use, a process that Gould and Vrba (1982: 5) defined as *cooptation*.³ Cattani’s (2006) in-depth analysis of how Corning’s pioneering research in fiber optics mainly grew out of its long-standing expertise in specialty glass provides evidence on the role of exaptation as a source of novelty, specifically the appearance of a new radical technology and its impact on the evolution of the telecommunications industry. In this sense, exaptation represents an important condition for a speciation event to occur. Exposing its dynamics means unearthing the origins of novelty, including radical innovation.

2.2 Exaptation and capabilities: luck and foresight

How firms develop and sustain competitive advantage in the face of technological change is a core question in strategy research. The issue is especially important in R&D-intensive industries, where superior performance depends on being consistently innovative over time (e.g., Nelson 1991; Teece *et al.*, 1997; Helfat, 2007). Typically, strategy research traces firm abilities to innovate back to the heterogeneity in firm resource endowments (e.g., Rumelt, 1984; Barney, 1991; Peteraf, 1993). To date, however, it has not systematically investigated whether these differences are the result of strategic foresight or historical accidents (luck). A firm’s history, for example, may endow it with knowledge for reasons unrelated to how the knowledge is ultimately applied in a new area of opportunity. To derive

3 Exaptation can also be observed when a character whose origin cannot be ascribed to the direct action of natural selection (a non-aptation), is co-opted for a current use—cooptation (Gould and Vrba, 1982: 5).

normative implications of strategy research findings, it is important to understand whether differences in resource endowments reflect luck or strategic foresight.

Strategy researchers have developed theory concerning firm heterogeneity and whether it is the result of luck or strategic foresight (e.g., Barney, 1986, 1997; Garud *et al.*, 1997; Cockburn *et al.*, 2000; Denrell, 2004; Cattani, 2005, 2006). A few empirical studies have also examined the conditions that lead to differences in firm performance. This work highlights the positive impact of prior experience (existing knowledge) on innovation and firm performance, especially when firms re-deploy the knowledge they gained in one arena into a new application domain (e.g., Carroll *et al.*, 1996; Holbrook *et al.*, 2000; Klepper and Simons, 2000; Helfat and Lieberman, 2002; Cattani, 2005). Such research has elucidated the relationship between prior experience, innovation, and firm performance by clarifying *why* and *when* variation in prior experience can be a critical initial condition. What is not clear is the extent to which the ability to innovate in a new domain is a consequence of a firm recognizing the value of its experience in other domains. In other words, do firms innovate because they anticipate which knowledge will be needed in a new domain, or is it the environment that randomly selects firms whose knowledge matches the requirements of a new domain?

One side of the debate argues that if superior performance results from differences in initial resources distributed randomly “at birth,” some firms will outperform others “merely because the environment happens to change in such a way that it favors their particular resource environment” (Henderson, 2000: 286). The other side argues that competitive advantage results from “any kind of managerial foresight or strategic insight” (Henderson, 2000: 289). As both luck and foresight can play a role, potentially, the research challenge is to clarify when each of these forces is at work and where and how it plays a significant role.

The precise role of luck and foresight in this process of technological innovation is difficult to determine precisely. However, the notion of exaptation suggests that in the course of technological evolution, there is a “dividing line” between (1) the phase in which a firm accumulates knowledge without anticipating subsequent applications and (2) the phase in which the same firm intentionally develops and leverages that knowledge in order to apply it in a new domain. It is at this critical juncture—when an intersecting relationship between a firm’s existing technological knowledge base and a new user environment is perceived and a functional shift occurs—that higher levels of foresight can guide the search for new technological applications. Understanding firm heterogeneity, therefore, must explain how and why certain firms accommodate while certain others resist a functional change.

It is a well-known fact, however, that a firm’s existing knowledge base constrains its opportunities and, by implication, the growth paths it can undertake. These constraints may involve managers themselves because they “undeniably lack perfect foresight” and hence their “decision-making heuristics and cognitive biases may lead them to make mistakes” (Helfat, 2007: 119). One such mistake is to overlook that the range of possible applications for a firm’s knowledge base is typically wider than its current applications. Firms can in fact capitalize on previous investments by transferring knowledge already available in-house to new applications, a capability that Garud and Nayyar (1994) have defined as transformative capacity. This capacity implies the existence of organizational practices and routines that are intended to preserve the integrity of the firm’s knowledge base and enhance its ability to seize novel opportunities as they come along. For example, a well-established practice at Corning was to ask senior researchers who were well-known in the scientific community and had a wide knowledge of Corning’s technical skills to act as Corning’s R&D ambassadors (Cattani, 2006). In this role, they were expected to visit research laboratories and existing or potential customers, attend conferences, etc. The main objective of having “technology scouts” was to spur innovation by facilitating the acquisition and diffusion of knowledge, both technical and at the market level. This practice permeated Corning’s efforts as it consistently searched for applications for which it had potentially valuable knowledge. New technologies (e.g., fiber optics) emerged as result of an inherently exaptive process, whereby existing technologies (e.g., specialty glass for cooking purposes) were eventually co-opted into new domains of application (e.g., telecommunications), which involved a major functional shift.

The previous example is illuminating because it shows how firms’ ability to capitalize on technological opportunities can be enhanced via organizational design and ad hoc routines and practices that may help firms to organize or prepare for seemingly serendipitous discoveries (Garud *et al.*, 1997; Dew *et al.*, 2004). So, even when firms cannot anticipate new applications, especially those that entail a major functional shift, some foresight might still be embedded in the process by which new technologies are created, more precisely the way the process is actually organized and managed, regardless of the final outcome (Cattani, 2008). The extent to which existing technological knowledge may prove functionally valuable for alternative applications—and hence exhibit potential for exaptation—is

also contingent on the specific features of that knowledge, including the “modular” nature of a technology (see next paragraph).

2.3 Modular exaptation

The degree of modularity of a technological system (Simon, 1962) affects exaptation in a static and dynamic sense (Andriani and Carignani, 2014). First, modularity offers the opportunity to classify different exaptation types based on the number of nested levels on which the exaptation process operates. The simplest case occurs when a new function emerges for an entire artifact and the artifact is selected by users for its new function. This case involves perfect morphological continuity. This type of exaptation is very common in the pharmaceutical sector, in which drugs developed for a specific disease are prescribed for several other diseases (DeMonaco *et al.*, 2006), but relatively rare in other sectors. A more complex case is when a module of a larger system undergoes a functional shift. The new function attribution may engender the separation of the module from its existing architecture, its transfer to a new domain and the emergence of a new architecture built around the transferred module. The third type of exaptation occurs when the module undergoing a functional shift remains in the original modular architecture but with a changed function. The functional change causes a reconfiguration of the internal architecture contributing to an increase in the efficiency level of the overall architecture, but not to an overall functional change.

The extent to which exaptation is responsible for radical innovation depends on the novelty of the functional shift (Andriani and Cohen, 2013). A function may be new with respect to the technology that performs it. For instance, gin was exapted from a drink used to alleviate circulatory problems to an intoxicating liquor (Burkhart, 2010); the function of intoxication by alcohol was well-known and plenty of other options were available (Schivelbusch, 1992). Or a function may be new to the world, like antisepsis, which represented a radically new medical capability. The former case corresponds to niche *preadaptation* (relative novelty): the exapted artifact is preadapted to an existing niche and no (or only minimal) modifications are necessary before its market introduction. Most off-label prescriptions in the pharmaceutical industry are an example (DeMonaco *et al.*, 2006). The latter case corresponds to niche *construction* (absolute novelty): the exapted component requires a long, adaptive process. For example, the first microwave oven appeared in 1947 after the exaptation of the magnetron in 1945. However, it took nearly 20 years to elaborate the features that defined the microwave oven market niche (Osepchuk, 1984).

The dynamic aspect refers to the fact that modularization of an architecture unfreezes the modules from the tight constraints of the architecture's interdependencies (Andriani and Carignani, 2014), thereby generating *option value* (Baldwin and Clark, 2000) for the modules that become available for recombination and hence exaptation. The literature on modularity has focused on recombination but so far has neglected exaptation.

2.4 Effectuation

On the premise that management is the art of dealing effectively with the reality of bounded rationality in a changing environment, Dew *et al.* (2008: 38) propose a model of entrepreneurial firm behavior that “emphasizes transforming current realities to fabricate new environments rather than acting within extant environments.” Underlying the model is what the authors call effectual logic—which is embedded in a theory of the world that refutes clear goals, accurate predictions, and control over outcomes. Effectuation is a logic of design rather than a logic of decision or discovery. Central to effectuation's logic is the rejection of prediction and forecasting in favor of a creative and constructivist approach to problem solving. Effectuators do not merely view opportunities as given or outside of their control but instead work to fabricate, as well as recognize and discover opportunities. As Sarasvathy (2001, 2008) pointed out, the effectual logic can be juxtaposed to the causal logic which is premised on the belief that better predictions lead to better control over outcomes. By not taking “the environment as given and/or predictable also implies that adaptive or other types of reactive strategies are inadequate and even inappropriate in the effectual process” (Dew *et al.*, 2008: 43). One of the key conceptual building blocks of the theory is that entrepreneurial firms have a predominately exaptive orientation and, therefore, adopt exaptive strategies to fabricate novelty (e.g., a new technology, market etc).

Exaptation is one of several types of transformations that entrepreneurs produce when employing effectual heuristics (Dew *et al.*, 2008). Exaptation transforms resources by converting them from established uses (things they were designed for) to new uses for which they were not originally designed. Burgelman's (1983) distinction between induced and autonomous initiatives is relevant in this light. While induced initiatives fit in the existing categories

used in the firm's strategic planning and take place in relationship to its familiar external environments, autonomous initiatives provide the basis for extending the frontiers of corporate capabilities and for the discovery of additional synergies among the existing resource combinations. Autonomous strategic behavior, in other words, is a crucial means for elaborating and exploring the potential applications of a firm's technological knowledge base and even the creation of new markets or industries.⁴ Firms can proactively create organizational conditions to foster autonomous initiatives (e.g., 3M's rule that employees can use 15% of their work time in the pursuit of new ideas) that in turn increase the likelihood of innovation by exaptation. The type of initiatives that will emerge and the criteria by which they are selected very much depend on the firm's internal selection mechanisms, such as those regulating resource allocation decisions. Firms cannot establish beforehand the specific outcome that these mechanisms will be able to generate; however, they can try to modify them to reflect more closely a change in the external environment and/or new strategic priorities. In this sense, resource allocation processes function as sorting mechanisms by which new ideas, investment opportunities, or techniques are eventually selected in or out.

Exaptation as a force of origination, important as it is, by itself cannot fully explain technological change and economic evolution (in this issue Dew and Sarasvathy, 2016). This requires the integration of exaptation and niche construction theories into a coherent conceptual framework that looks at the dynamic interplay between these two critical evolutionary forces.

3. An integrated model of the genesis of exaptation

In this section, we present an initial attempt to model the emergence of novelty—and particularly new market niches—triggered by the discovery of new functions for existing artifacts. The development of market niches as a result of exaptation differs from that based on adaptive innovation, in at least three important respects:

1. Technological continuity: the core of the new technology is transferred across (usually) unrelated markets and does not have to be designed from scratch.
2. Creativity: the origin of the new idea is rooted in hidden functionalities of existing artifacts, whose discovery originates from artifact–agent interactions that occur within dense social networks.
3. Selection criteria: in the case of exaptive functional novelty, the selection criteria of a new market niche do not exist and have to be developed.

The model—which is summarized in Figure 1—identifies two main processes: a process that we call functional shift because it leads to the emergence of a new function for an existing artifact; and a process resulting in a functioning market niche. We start by arguing that the discovery of new functions is not purely driven by chance, but that three concepts explored in this special section help explicate what Garud *et al.* (2016) in this issue call *manufactured luck*. These concepts are (i) narratives as performative acts, (ii) affordances, and (iii) functional fixedness. The action of these forces may be reinforced by environmental factors such as exogenous shocks. The emergence of a new function, however, is only the first necessary—though not sufficient—condition for the creation of a market niche. The novel, often inchoate, function has to be transformed into a viable economic market niche. As Dew and Sarasvathy (2016) claim, the literature on exaptation has mostly focused on proving that exaptation actually occurs and is a critical source of novelty. However, relatively less attention has been devoted to studying the processes that turn a new function into a new market niche.

3.1 Functional shift

What does trigger functional shift? To approach this question we need to clarify two aspects: function and the relationship between function and artifact. We define a function as the reason for which an artifact is selected (De Winter, 2010).⁵ This implies that, for instance, the same artifacts can be used for different functions in different market niches. Because no one can anticipate the number of ways an artifact will be put into use, the number of functions

- 4 This process of generating new technologies, and quite possibly new businesses, from the same base of experience is very much like the process of speciation by lineage splitting in biology.
- 5 Note that this definition, in line with the etiological analysis of function, does not involve the intentionality of designers but only the selection criteria of the users.

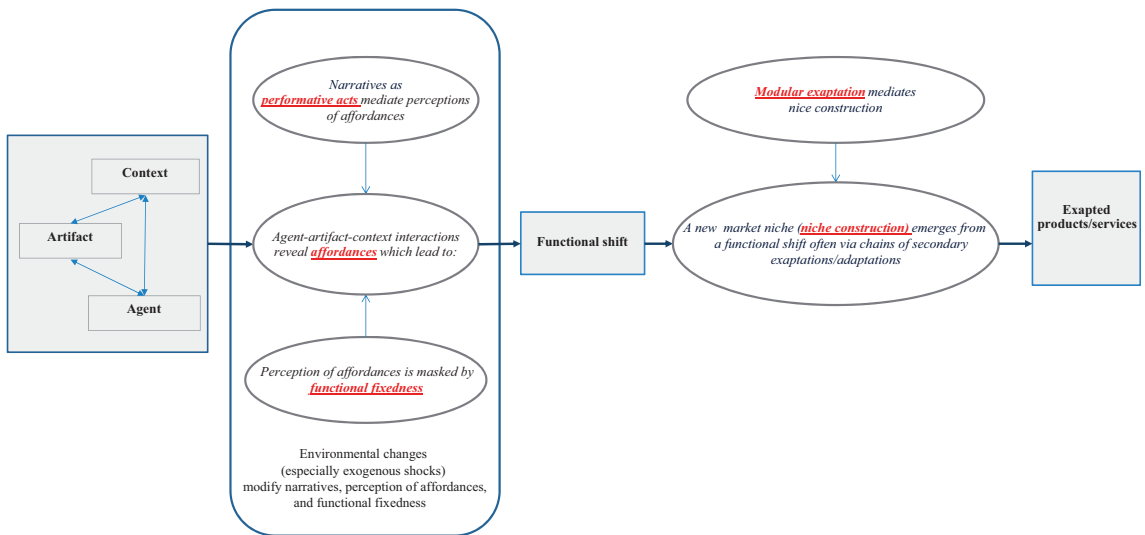


Figure 1. An integrated model of exaptation.

of an artifact is un-prestable (Kauffman, 2000). The relationship between an artifact and the functions it enables is complex because each artifact is associated with two sets of functions. The first set includes historically determined functions that define the artifact, a subset of which is likely to have informed the design of the artifact itself. This set is finite and the functions that belong to it can be ranked according to a predefined set of criteria. The second set consists of potential (mostly exaptable) functions. The number of functions in this set is indefinite and no association with specific markets can be established ex ante. A functional shift is concerned with the second set.

How do functions emerge in the second set? Drawing from Gibson (1950), in this issue, Felin *et al.* (2016) propose the idea of “affordances.” As they argue, affordances “are the perceived possibilities that actors or organisms have related to objects and their environment.” From this perspective, functions are not a property of the artifact but instead express a relation between the artifact and other artifacts and/or agents within a specific context. Conceptualizing function as a relational property between different entities transforms it from an attribute of an entity (the artifact) to a set of relations linking the central artifact to an agent (and possibly to other artifacts and agents) within a shared context. If functions are relation sets, then a functional shift also involves a shift in existing relations. This can be restated as a change of context, in which an artifact is segregated from its relation set, “translated” from its context of origin (which we call the base domain) into a new context (which we call target domain) and then re-associated with a new set of relations.

An analysis of historical cases of functional shifts as reported in extant literature reveals that such a transfer entails considerable creativity and imagination, albeit within the constraint of artifact continuity. As Felin *et al.* (2016) in this issue argue, functional shifts are not the result of systematic experimentation with possible target domains because such domains are nearly infinite, thereby denying the possibility of a systematic approach. This implies that framing functional shift in terms of analogy, in which the base and the target domains preexist, does not help to solve the problem of functional shift. The perception of functional shifts relies on a creative effort that consists in imagining a target domain for the artifact that does not yet exist. The antiseptics case discussed before clarifies this point. Lister’s antiseptics method derived from an intuition concerning a shared attribute between two apparently entirely unrelated situations, that is, the similarity between odors from putrefaction in sewers and odors in suppuring open wounds. From there he hypothesized that putrefaction and suppuration share the same cause (germs). Lister effectively generated a target domain for the use of carbolic acid as a preventative tool. The emergence of a target domain for carbolic acid, therefore, did not precede the functional shift of carbolic acid (had that been the case, then Lister would have embarked on a systematic search for potential antiseptics). In this case like in other such cases, the functional shift coincides with the emergence of a target domain. The association between deodorization and antiseptics was the result of a confluence of various factors: first, the belief in the causal role of germs in infection and second

the affordance that carbolic acid offered with respect to germ control (even if originally in an entirely unrelated field). It is important to emphasize that such affordance played an active role in pushing carbolic acid from deodorization to antiseptis. Although the theory of germs was a necessary condition for antiseptis, it was the affordance of carbolic acid that created modern antiseptis. Agency, therefore, was distributed between artifacts, agents, and existing knowledge.

The antiseptis example sheds light on the role of affordances in the emergence of novelty. What this and other such cases seem to suggest is that affordances lead to new functionalities in the economy separately from traditional drivers of innovation, that is, pull from new needs or push from the discovery of new phenomena. As [Felin et al. \(2016\)](#) note:

The condition for an exaptation to occur does not necessarily lie in what the actors, basing on their knowledge-base and “experience,” transfer from one domain to another. In other words, new creative uses and exaptations also emerge as a result of affordances conceived of by organisms—and do not only depend on past experience.

The concept of affordances identifies three main forces driving innovation via exaptation: agents, artifacts, and environments. It is the interaction among these three forces that fosters the emergence of radical innovations such as antiseptis, microwave ovens, or fiber optics. Focusing on the interaction among artifacts, agents, and environments, however, also raises some questions. If artifacts have the potential of yielding new functionalities and the number of new functionalities is un-prestatable, then why has the role of exaptation gone almost unnoticed in most accounts of technology and innovation? We believe this is partly a problem of framing. Lacking a theory and, by implication, a clear understanding of exaptation, many exaptive events have been ignored or reported under different headings (e.g., serendipity, luck). But it also partly reflects an anthropological difficulty in dealing with functional shift in the absence of a morphological change. Cognitive psychology has identified a number of bias that shape attention and creative thinking ([Kahneman, 2011](#)). In particular, *functional fixedness* plays a crucial role in funneling creativity along avenues that favor morphological changes over functional changes, in other words form follow function and not vice versa ([Felin et al., 2016](#)). Functional fixedness derives from a conflation of form and function, in which a dominant cognitive association is established between a form and its archetypical function. If an artifact is “collapsed” into a strict function–form association instead of being perceptually seen as a bundle of affordances, explorations based on functional shifts are *de facto* cognitively excluded. A number of fascinating hypotheses about the role of functional fixedness in entrepreneurship and organizational change can be explored in future research.

If a functional shift involves establishing a new linkage between material objects, agents, and contexts that were previously unconnected, and if existing analogies are of limited use because target domains emerge through functional shifts, then the question is what type of cognitive approaches is more appropriate. [Felin et al. \(2016\)](#) criticize the idea of perception as objective representation of reality and propose embodied cognition as a promising approach to frame functional shift. [Garud et al. \(2016\)](#) suggest a complementary approach to address the following questions:

How might we conceptualize exaptation in the context of technological change in a way that facets of its use in evolutionary biology are preserved and yet human agency to “foresee” exaptive possibilities serendipitously are acknowledged? In other words, how might one go beyond blind luck that representation still suggests, to also induce exaptation without knowing the specific future uses of a technology?

Drawing from *actor-network theory* ([Latour, 2005](#)) and its controversial claim that agency is distributed through material and living entities, [Garud et al. \(2016\)](#) argue that narratives constitute relationships between social and material elements. While purely descriptive narratives consider the interaction between features (e.g., artifacts) and contexts as essentially exogenous, the relational facet of narratives highlights a capacity for forging such connections. By elaborating narratives, actors (e.g., entrepreneurs, managers) give meaning and functionality to what they encounter or experience, forge new feature–context connections, and so unveil new affordances. The dimension of relationality introduced by narratives aptly fits with the conceptualization of functional shifts as changes in networks of interactions between artifacts, agents, and contexts. It also delineates a series of promising research directions that we will describe in the following section.

The three factors described above—affordances, narratives, and functional fixedness—operate in a broader environment which may facilitate or hinder functional shifts. Exogenous shocks constitute a particular interesting case of rapid environmental change. Exogenous shocks change the context in which companies and technologies operate in terms of, among others, regulatory frame, nature of interdependencies among the actors of a socio-technical network, procedures and routines, etc. By perturbing the status quo, exogenous shocks may enable functional shifts by revealing affordances, requiring different narratives, and breaking the gridlock of functional fixedness. Several events of historical significance have been ascribed to exogenous shocks such as the appearance of agriculture due to the desertification shock—the so-called Natufian hypothesis (Balter, 2010)—the emergence of cities (Brooks, 2006), radical transformation of industrial clusters (Bathelt and Boggs, 2003; Best, 2003). However, the most frequently invoked explanatory mechanism for system reconfiguration is recombination rather than exaptation. Functional shift is rarely, if ever, considered. We advance the hypothesis that a functional shift works in parallel with recombination by making new functional components available for recombination. In this issue the paper by Ching (2016) illustrates the importance of exaptation in this context. Ching studies a transformation of the Internet video industry in China following a major, sudden, and unexpected, regulatory change. He finds a surprising result: firms that adapted and changed their user-generated content (UGC) business model to the eventual dominant professionally produced content (PPC) business model performed better than firms that started with the dominant model. Ching convincingly shows that companies burdened with a legacy business model in a radically changed regulatory environment nevertheless exapted features of the old business model that allowed them to compete and dominate against companies “blessed” by the regulatory change. In the language of our model, the regulatory change revealed affordances but, at the same time, broke the functional fixedness that rigidly separated the pre-regulatory-change UGC model from the post-regulatory-change PPC model.

3.2 The emergence of niches

Until this point, we have discussed the emergence of functional shifts. This raises the question of whether functional shifts directly impact the economy by creating new economic demand. To address this question, in this issue Dew and Sarasvathy (2016) draw from an new frame in evolutionary biology called niche construction (Odling-Smee *et al.*, 2003) to expound on the relation between exaptation and the creation of new market niches. Building on previous work that links adaptive and exaptive processes such as *exaptive bootstrapping* (Lane, 2011) and the *design-exaptation cycle* (Andriani and Carignani, 2014), Dew and Sarasvathy notice that “to understand why (and when) exaptations have an economic impact we need to trace complementary behavioral processes that leverage the initial opportunity an exaptation provides. And, since an initial exaptation may become consequential only via follow-on niche construction processes that exploit the opportunity it provides, understanding these processes may be key to understanding the economic impact of exaptations.”

In the example of antisepsis, the early 1865 experiments were followed by several years of work in which the architecture of the new niche was defined in terms of elaboration of surgical procedures and routines, identification of complementary technologies such as types of bandages and sutures, definition of labels that then generated a community of like-minded surgeons that help to spread the antisepsis message, emergence of industry that would supply the necessary materials and appropriate surgical tools, inclusion of the antisepsis theory and practice in the curriculum of medicine universities and textbooks, etc. In short, it took more than a decade before the selection processes that define a correctly operating niche were in place. Similarly, even though Corning invented the first low-loss optical glass fiber for long-distance telecommunications applications in 1970, a sizeable market for optical fibers did not emerge until the early 1980s (Cattani, 2006).

Dew and Sarasvathy argue that the niche construction process falls in between two extremes: the case in which economic demand (at least in its embryonic form) exists from day one (e.g., the anti-tuberculosis turned overnight in the first anti-depressant), and the case in which all the elements of a niche must be created almost from scratch (e.g., antisepsis or Viagra). Even in the first case, however, building legitimacy for the new anti-depressant niche required years of performative action. Linking exaptation to niche construction not only clarifies the dynamics of exaptation as a net contributor to novelty, but also calls for a more balanced approach in evolutionary theory, one that puts external forces (selection) and internal forces (e.g., niche construction arising from exaptation) on the same level. The need to modify evolutionary theory as typically applied in economics and innovation studies can be traced to the selection assumption. It is commonly assumed that selection drives the process of evolution of new niches by

screening mutations produced by economic actors. But this implies the preexistence of a fitness function by which selection can discriminate among alternative options. Such a function, in turn, depends on the existence of the niche itself, that is, the set of specific and relatively fixed interdependences among artifacts, agents, and contextual features that define how economic goods are produced, consumed, and priced. As exaptation can generate new functions and market niches, it follows that selection criteria are formed during the process of niche construction and not the other way round. The paper by Dew and Sarasvathy starts tackling the problem of exposing the mechanisms underlying exaptation-based niche construction.

4. Where do we go from here

Integrating the previous insights and possible extensions illuminates a number of promising avenues for studying exaptation as a key source of novelty in the social sciences.

4.1 Organizational memory and exaptation

Firms can enhance their ability to produce novelty by deploying their knowledge into new applications. For this to be possible, however, knowledge must be first stored or retained in the organization's memory (e.g., [Argote, 1999](#)). Organizational memory is not stored in only one location but distributed across various knowledge repositories or bins, such as an organization's employees, roles and structures, standard operating procedures, and so on ([Walsh and Ungson, 1991](#)). How effectively knowledge can be retained depends on its properties. Knowledge properties significantly affect "the rate at which knowledge is accumulated, how much of it is retained, where it is retained, and how easily it diffuses within and across firm boundaries" ([Argote et al., 2003: 574](#)). For example, it is harder to retain and transfer tacit and difficult-to-articulate knowledge ([Polanyi, 1962](#); [Winter, 1987](#)) unless it is codified or not kept in written format.

The problem is even more complex in the presence of high turnover rates among organizational participants, especially when organizations are low in structure—i.e., work cannot be standardized or routines are not available to handle the loss and addition of participants. While organizational memory does not necessarily coincide with individual memory, individuals are often "the sole storage point of knowledge that is both idiosyncratic and of great importance to the organization" ([Nelson and Winter, 1982: 115](#)). The departure of long-tenured individuals or individual with highly specific skills and expertise might, in fact, amount to loss of (tacit) knowledge with obvious consequences for organizations. The challenge for the management of many organizations is not simply to replace individuals with years of experience with other with similar skills and expertise, but to retain or recreate "the sophisticated, context-dependent knowledge that resides with the employee who is leaving" ([DeLong, 2004: 19](#)).⁶

Preserving an organization's memory is critical for facilitating the emergence of novelty via exaptation. In their study of IDEO's way of organizing for innovation, for instance, [Hargadon and Sutton \(1997\)](#) emphasized how the process of storing, retaining, and retrieving knowledge enabled IDEO to find new applications for its technologies (artifacts), both those currently in use and those with no obvious application. Interestingly, the idea that the same technology or artifact may undergo a major functional shift was implicit in the company's search activities and, in general, *modus operandi*. Technical problems tend to change continuously, but the same body of knowledge can remain valuable as individuals "recognize similarities between old solutions and new problems" ([Hargadon and Sutton, 1997: 732](#)). The extent to which an organization's memory can be considered a prerequisite for exaptation is a very important question that future research might find it worthy of further exploration.

- 6 Some have pointed out that it would be challenging for the US space agency – the NASA – to send astronauts back to the moon again. A former NASA manager noted: "If we want to go to the moon again, we'll be starting from scratch because all of that knowledge has disappeared. It would take at least as long and cost at least as much to go back" (reported in [DeLong, 2004: 11–12](#)). That is because "in the 1990s NASA lost the knowledge it had developed to send astronauts to the moon. In an era of cost-cutting and downsizing, the engineers who designed the huge Saturn 5 rocket used to launch the lunar landing craft were encouraged to take early retirement from the space program. With them went years of experience and expertise about the design trade-offs that had been made in building the Saturn rockets. Also lost were what appear to be the last set of critical blueprints for the Saturn booster, which was the only rocker ever built with enough thrust to launch a manned lunar payload" ([DeLong, 2004: 11](#)).

4.2 Exaptation as a shadow option

The previous discussion about affordances suggests contact points with a real-option perspective on organizational resources (Bowman and Hurry, 1993; McGrath, 1997), particularly the notion of a shadow option. Specifically, a shadow option refers to the case in which an existing bundle of resources provide options (or investment opportunities) awaiting recognition. Recognition of a shadow option entails retrospective sense-making. This occurs when new information about possible applications for a firm's knowledge stock gradually emerges. The logic underlying a shadow option is distinct from the logic of other types of options whereby an initial upfront investment is deliberately made to achieve a specific end, such as an option to grow. In fact, opportunities for strategic choice come "into being only when decision makers recognize them" (Bowman and Hurry, 1993: 763). The recognition of a shadow option depends on whether an organization is capable of preserving its memory (as pointed out before), as well as institutionalizing routines and procedures that are explicitly designed to facilitate the identification of new opportunities.

Shadow options may be turned into real options when a functional shift occurs that unveils opportunities to further invest in existing assets. A functional shift then can be thought as a link between an asset that is the result of a previous investment and a future investment opportunity. Put it differently, an investment opportunity should be evaluated not only for its real option value but also for its shadow option value. This raises a fundamental question: is it possible, at least in principle, to anticipate the shadow option returns (the exaptive returns) deriving from a real option investment? Although the un-prestateability argument prevents the calculation of a probability distribution, historical data may provide some general guidelines. For instance, in the pharmaceutical industry we know that investment in drug development is likely to generate additional uses for the same drug at zero cost or low cost (DeMonaco *et al.*, 2006), some of which may become blockbusters (e.g., Viagra). Even though the returns for shadow options uses depend on several factors, some of which are stochastic, future research may explore how such returns can be inferred and so establish a tighter link between shadow and real options.

4.3 Exaptation as a driver of user and open-source innovation

Several studies demonstrate that new technological trajectories, markets, or industries often result from linking different knowledge from across disparate actors, communities, or social networks. The literature on users' innovation (Von Hippel, 2005), for instance, has demonstrated how users represent a fundamental source of innovation because they manage to conjugate very diverse contexts with a very large number of experiments (Von Hippel and De Jong, 2010). A fascinating example of the importance of users in the modern economy is provided by the Makers movement (Anderson, 2012). The Makerspace is made of a library of digital objects sitting in the Commons⁷ that makers can turn into physical objects by means of 3D printers.⁸ The Makerspace is characterized by modularity, democratized access, and combinatorial rate of growth provided by the standardized interfaces of CAD programs, the low (entry barrier) costs of acquiring 3D technologies, and the nearly infinite combinability of digital objects. Turning the library's digital objects in physical objects via 3D technologies generates the possibility of exapting digital objects by exposing their physical forms to new unforeseen contexts. In other words, the Makerspace facilitates the interaction between a nearly infinite library of digital objects and the practical context of their applications. Because they allow for the complete separation between design and manufacturing, 3D technologies democratize access to both design and manufacturing. Individual users can now control all the phases from conceptual design to manufacturing. Localizing design and manufacturing under the roof of the same metaphorical "garage" enables users to expose digital and physical objects to new contexts, discover hidden functions, and then activate the design-exaptation cycle. Future research may further explore the conditions under which the increasingly larger number of users operating in the highly modular environment of 3D technologies can enhance innovation by fostering exaptive processes.

4.4 Exaptation and the emergence of new technological paradigms

In a number of documented cases, exaptation has been found to be responsible for the generation of radical innovations that have spurred the emergence of novel trajectories and paradigms. As noted before, the introduction of anti-sepsis marked a revolutionary advancement in the field of medicine that not only revolutionized the practice of

7 The term Commons refers to the cultural and natural resources that can be accessed without discrimination by all members of a society.

8 Or, the other way round, from objects to digital objects by means of 3D scanners.

surgery but also evolved into a new paradigm of infection prevention in surgery. Typically, paradigms are embedded in communities of scientists and technologists that form around shared interpretations of available evidence and theories, as well as expectations about what constitutes legitimate future research in both the scientific and the technological domains. The establishment of a technological paradigm implies the formation of a consensus around the directions of technical change to pursue and those to ignore. As Dosi (1982: 153) aptly pointed out:

Technological paradigms have a powerful exclusion effect: the efforts and the technological imagination of engineers and of the organizations they are in are focussed in rather precise directions while they are, so speak, “blind” with respect to other technological possibilities. At the same time, technological paradigms define also the idea of “progress.”

Breaking an existing paradigm requires an accumulation of anomalies concerning both fundamental theoretical and experimental aspects (Kuhn, 1962). We advance the hypothesis that exaptations may play a role in this respect. Evidence shows that exaptations reveal possibilities that sit outside the envelope of paradigm-based legitimate science and technology, thereby questioning the exclusivity of interpretation of the body of available evidence. The emergence of biological psychiatry proves this point. In the 1950s, the quick appearance of exaptations concerning antidepressants (Kline, 1970), tranquilizers (Tone, 2008), and anti-psychotic drugs (Raviña, 2011) unveiled a novel and unanticipated dimension for pharmacological application, that is, the treatment of mental disorders. This new evidence *de facto* called into question the legitimacy of psycho-analysis as the only approach for treating those disorders. By the end of the decade, biological psychiatry had replaced psycho-analysis as the dominant paradigm. As this example suggests, the relationship between exaptation and paradigm constitutes a very promising, though still unexplored, field of inquiry.

4.5 Methodological challenges: from longitudinal studies to quantification

One of the main challenges for studying exaptation is to identify appropriate—and possibly widely shared—methodological approaches. A simulation approach has been used to examine the dynamics of exaptation. For instance, Villani *et al.* (2007) proposed an agent-based model that describes the origins of innovation by focusing on the interaction between artifact innovation and attributions of functionality. Similarly, Winter *et al.* (2007) developed a model based on fractal geometry to examine the dynamic interplay of luck, strategic foresight, and market feedback. By allowing examination of the relative importance of these evolutionary forces, the model sheds light on the logic of the early stages of a new market or industry in which participating firms are diverse in their relevant backgrounds and capabilities. Although formal modeling has been used to describe the dynamic of exaptation, researchers still face the challenge of devising ad hoc research strategies for documenting and analyzing *when* and *how* exaptation occurs in real-worlds situations.

Some scholars have adopted a longitudinal case study design to empirically expose the dynamics of exaptation. Cattani (2006), for instance, conducted an in-depth historical analysis of the micro-organizational processes and evolutionary forces (internal and external to the firm) responsible for the emergence of fiber optics technology. Longitudinal case studies that rely on a variety of data and particularly original documents detailing the decisions and actions of the actors involved in developing a new technology, precisely when those decisions and actions were first taken, constitute a very promising research approach. In general, detailed historical data might be used to identify key shifts in the environment (e.g., an exogenous regulatory change) that produce exaptive functional changes in existing artifacts (e.g., a technology) and their impact on the evolution of a nascent market or industry (e.g., Ching, 2016). For studies interested in the emergence of new technologies, it is critical to chart *technological* entry, defined as innovation in a new technological domain, as distinct from *market* entry, which refers to initial production of a product or provision of a service (Malerba and Orsenigo, 1999; Helfat and Lieberman, 2002). To this end, patent data are particularly useful because they provide a detailed and consistent chronology of when certain knowledge was originally created, and can thus be used along with other sources (field interview data, industry and company reports, specialized books, and business press articles) to measure technological entry (Cattani, 2005).

But alternative approaches might just as well serve the purpose of studying exaptation. Recent advancements in textual analysis seem to be especially suitable for the articulation of a narrative perspective on exaptation. Textual analysis, in fact, would enable researchers to trace emerging narratives and their role in inducing technological exaptation (Garud *et al.*, 2016). Textual analysis could also be extended to studying the relation between exaptation and effectuation, particularly how effectuators fabricate, but also recognize and discover opportunities (Dew and

Sarasvathy, 2016); or how individual perceptions result in the identification of novel affordances—new uses or functions (Felin *et al.*, 2016). Lab experiments have yet to be used to study exaptation. The cognitive processes (e.g., the role of analogical thinking) involved might be better investigated in a lab setting where other intervening factors can be controlled.

But we also need studies that quantify the phenomenon of exaptation. Important questions to ask include: What percentage of innovations is due to exaptation? What is the actual contribution of exaptation to radical innovation? Is the exaptation process more relevant in certain industries than in others? Quantifying exaptation, however, poses several challenges: it requires the matching of a population of artifacts against a population of functions. However, the former is easier to determine than the latter. There are no databases about functions for most industries. Patent data could be employed to compute such a measure. For instance, the patent document, claims, and citations across different technological classes may be used as proxies for identifying functional shifts. The pharmaceutical sector is especially interesting in this regard because of the regulatory framework that assigns functions to molecular entities. By matching a population of drugs with the evolution of uses *before* and *after* drug approval, it would be possible to study the evolution of functions for a given set of artifacts (here drugs).

Of course, no single methodology is most appropriate for studying exaptation. It seems to us that multiple methods that cross disciplinary boundaries will be required to make progress in this area of research and theory. Mixing methods, criteria, and metrics from different disciplines is, indeed, where the scholarly opportunity lies. The greatest potential for advancing research and theory on exaptation lies in methodological innovations that can penetrate and describe the inherent complexity of exposing the micro-evolutionary processes underlying exaptation. In our opinion, this is indeed the greatest need.

5. Conclusions

The initial quote highlights the idea that many, if not most, things in nature were not created for the specific use they ended up performing goes as far back as to Latin poet Lucretius. Similarly, in the history of technology, exaptation is probably more common than in natural history: many of the most dramatic inventions of the modern age, in fact, were originally selected for quite different purposes than what eventually became their most enduring function (Mokyr, 1991). New technologies are rarely created from nothing but they usually originate from components that previously existed. In turn, these new technologies “offer themselves as possible components—building blocks—for the construction of further new technologies” (Arthur and Polak, 2006: 23). By exposing the process by which existing technologies become the building blocks for new ones, exaptation sheds new light on the evolutionary determinants of creativity, innovation, and diversity.

If exaptation is an important contributor to technological change and the emergence of new markets, then current evolutionary explanations of innovation that typically insist on adaptation are incomplete. We believe that filling this explanatory gap, which we provocatively call the “exaptive residual” (in Solow’s terms), may help resolve the puzzle of persistent and higher-than-average levels of creativity and innovativeness in dense and diverse social systems such as cities (Jacobs, 1969, 1985), clusters à la Silicon Valley (Saxenian, 1994), or research labs like the Bell Labs (Gertner, 2012). Although a careful reading of the literature reveals that functional change contributes to the innovation dynamics of these systems, surprisingly it remains hidden behind other explanatory mechanisms. To the best of our knowledge, the studies cited before and those in this special issue are the only studies so far that have explicitly tried to probe the role of exaptation and disentangle exaptive from adaptive processes in studying the emergence of novelty.

Increasingly, the concept of exaptation is viewed as a crucial source of novelty across a variety of disciplines; yet, its importance still has to be fully recognized. The main purpose of this special section is to introduce the concept of exaptation to a broader audience and expose its contribution to social scientists concerned with the origins of novelty, particularly radical innovation. We hope that this special section will stimulate researchers with diverse backgrounds to systematically engage in an interdisciplinary discussion on the theoretical and empirical implications of studying exaptation.

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